

ENGINEERING PORTFOLIO

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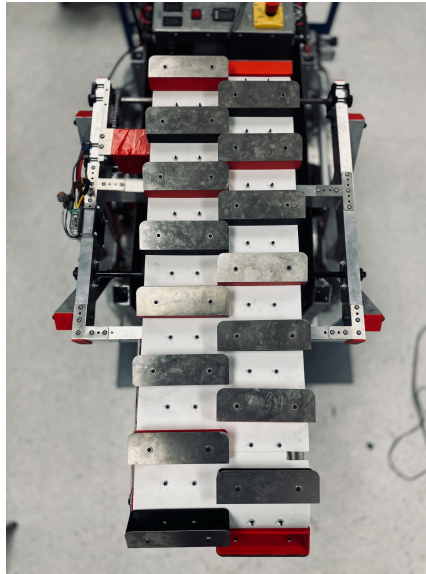
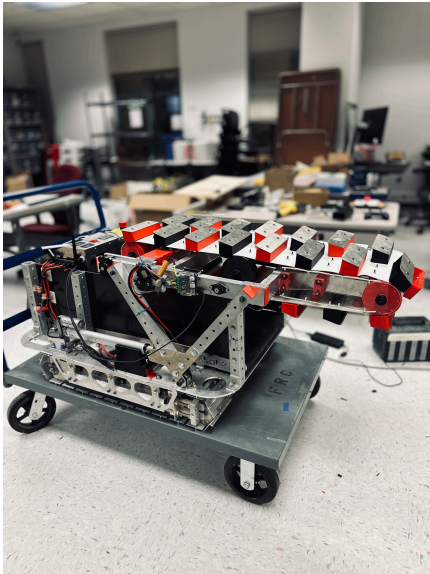
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NASA LUNABOTICS CHALLENGE

Mechanical Team Member @ CMU Moon Miners

Nov 2025 – Present



PROBLEM

Create an excavation system that maintains consistent regolith throughput despite unpredictable terrain interaction and strict subsystem coupling constraints.

METHOD

Explored multiple intake geometries and converged on a dual-channel architecture to decouple material flow from local terrain variance. Used simulation results to guide geometric tradeoffs between stiffness, manufacturability, and serviceability rather than purely maximizing safety factor.

RESULT

The final design demonstrated stable simulated flow behavior across varying load cases instead of peak-only optimization. Subsystem interfaces remained robust to alignment and mounting tolerance changes, reducing integration risk at the rover level.

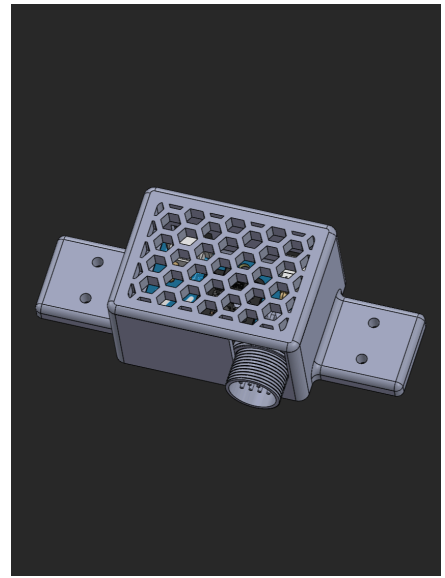
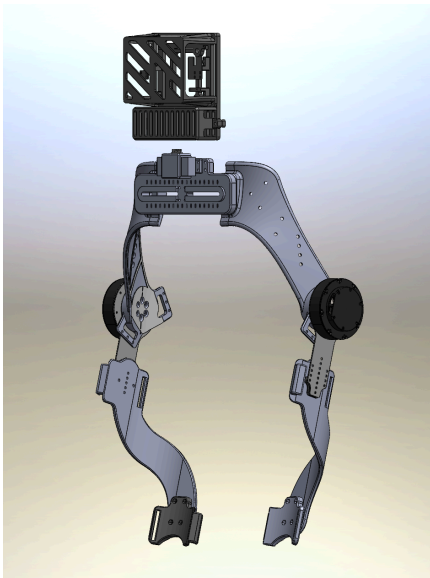
SKILLS USED

SolidWorks, ANSYS Discovery, Git, 3D Printing, Machining, Mastercam

HIP EXOSKELETON

Research Assistant @ CMU MetaMobility Lab

Sept 2025 – Present



PROBLEM

Improve assistive device effectiveness without increasing user fatigue or reducing long-term wearability.

METHOD

Redesigned mechanical interfaces to prioritize anatomical alignment and load path efficiency before actuator selection. Iterated hardware with fast prototyping to validate human-in-the-loop constraints that are not captured in simulation alone.

RESULT

Hardware revisions improved comfort and motion naturalness in addition to reducing mass. The platform became better suited for extended user trials and data-driven assistance development.

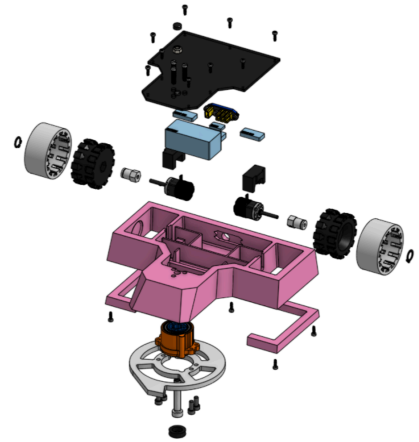
SKILLS USED

SolidWorks, Git, Python, Microcontrollers, 3D Printing, Soldering

NATIONAL HAVOC ROBOT LEAGUE

Team Member @ CMU Combat Robotics

Sept 2024 – Present



PROBLEM

Develop a weaponized robot architecture that remains controllable and survivable under asymmetric impacts and repeated shock loading.

METHOD

Designed the robot around energy management rather than raw weapon power, using compliance and mass distribution to protect critical components. Evaluated weapon geometry and inertia to shape impact impulse and spin-up behavior instead of maximizing tip speed alone.

RESULT

The robot maintained drivability and weapon consistency after high-energy collisions where rigid designs typically degrade. Design choices reduced failure modes related to shock transmission and post-impact instability.

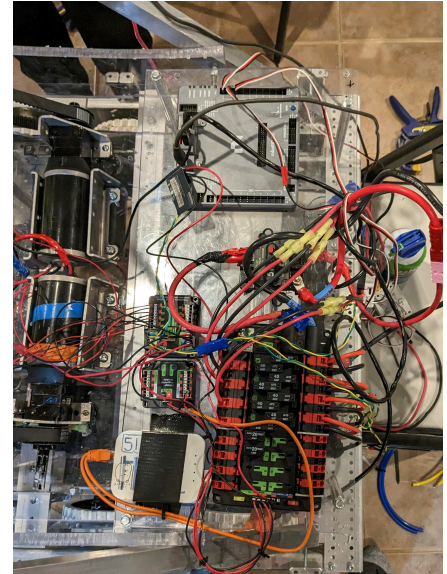
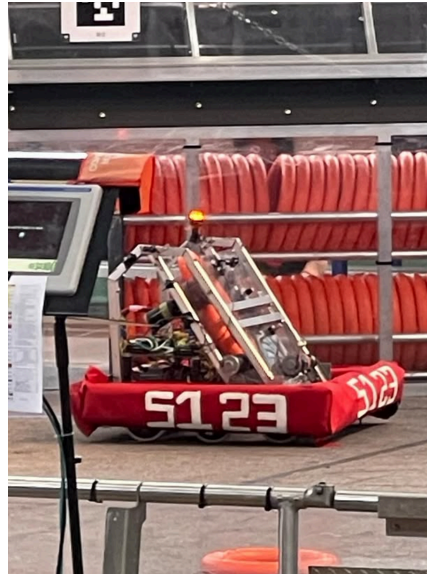
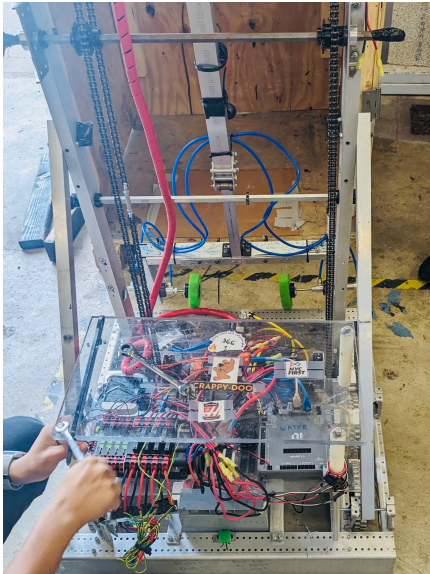
SKILLS USED

Onshape, MATLAB, 3D Printing, Machining, Soldering

FIRST ROBOTICS COMPETITION

Control Department Director @ FRC Team 5123

Oct 2021 – June 2024



PROBLEM

Enable consistent autonomous and teleoperated performance in an environment dominated by sensor noise, latency, and driver variability.

METHOD

Structured control software to isolate sensing, state estimation, and actuation, allowing independent tuning and rapid iteration. Treated vision as a probabilistic input rather than a ground truth signal and compensated for delay at the control level.

RESULT

Autonomous behavior became predictable and repeatable across matches rather than condition-dependent. Drivers reported increased confidence due to smoother control response and reduced correction workload.

SKILLS USED

Java, Git, Raspberry Pi, Soldering